# Many Outlets, One Story: Event-Level Evaluation and Re-ranking

**Group 1** · Uju Nwokeji (Data & Clustering) · Mehdi Hassan (Retrieval & Evaluation) · Olaniyi Moluga (Reranking & Analysis)

## Abstract

News search often returns many copies of one story, leaving the user with a single perspective on an event. Standard IR metrics such as nDCG do not penalize this redundancy. We build a time-scoped test collection of 170,781 English articles from Infini-News (CC-News), group near-duplicates into story clusters, and judge clusters (not articles) for 50 evaluation queries (plus 10 pilot). We compare four retrieval systems (BM25, bi-encoder, cross-encoder, hybrid) on nDCG and coverage metrics (α-nDCG, cluster recall) to quantify this gap, then measure when cheap rerankers (story cap, outlet cap, earliness prior) improve coverage without harming relevance. Judgments are validated with Cohen's κ.

## 1. Idea and Research Questions

**Goal:** A news search system should give comprehensive coverage of an event. Diversification is well studied for e-commerce search [3]; we study retrieval under news syndication, where copies of one story fill the ranking.

> **RQ1 (Measurement):** How much story redundancy do standard retrievers produce, and how much does nDCG overstate their coverage of an event?
>
> **RQ2 (Mitigation):** Do cheap rerankers improve coverage, and what do they cost in relevance?
>
> **RQ3 (Moderators):** How do query style (specific names vs. descriptions) and event size (number of relevant stories) change how well the rerankers work?

**Main Task:** The emphasis is strictly on evaluation and reranking. We are not building a new retriever; we measure how existing retrievers fail on news data and evaluate a lightweight fix.

**Why it matters:** In news monitoring, seeing the same Associated Press story 10 times across 10 outlets adds no information; users need different angles (the report, the reaction, the analysis). Metrics that ignore this reward systems for returning copies.

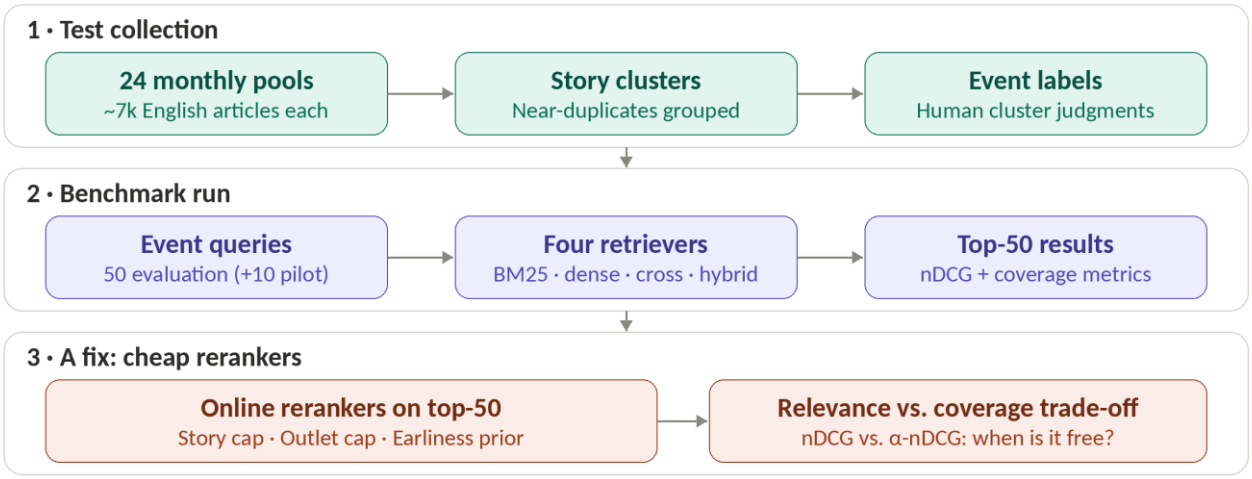



*Figure 1: Pipeline architecture: test collection → benchmark run → cheap rerankers.*

## 2. Data and Judgments

| Item | Value |
|---|---|
| Source / sample | Infini-News v2 (CC-News derived); 24 monthly shards, 2023-01 – 2024-12; 565,515 raw rows |
| Filtered English docs | 170,781 (~7,116 per monthly pool); mean length ~3,100 characters |
| Story clustering | Exact text_xxhash64 + MinHash/LSH (5-gram, Jaccard ≥ 0.8). 3-shard pilot (22,613 articles): 199 clusters with ≥5 copies, 180 spanning more than one outlet |
| Queries | 60 total (10 pilot, 50 evaluation) |
| Judgments | ~900 cluster decisions (~15 candidates × 60 queries: 750 evaluation + 150 pilot); 20% double-judged |

**Clustering:** Articles are grouped only within the same crawl day; an annotator can merge edited copies using a same_as field, and all metrics use the merged groups.

**Relevance and pooling:** Relevance is defined at the event level (which stories report the queried event) and judged per cluster, one decision per near-duplicate group: 2 = same event, 1 = related but different development, 0 = different event. Grade 1 is mapped to non-relevant in the main metrics and reported as an ambiguity rate in a sensitivity analysis. All systems are run before judging; candidates are the clusters in the top-10 of each system (S1–S4), deduplicated and capped at 15 by best rank across systems. Unjudged clusters count as non-relevant. Judging clusters needs ~750 evaluation decisions instead of ~356,000 article decisions (50 queries × ~7,116 articles).

## 3. Methods and Models

| ID | System | Description |
|---|---|---|
| S1 | BM25 | Classic sparse lexical baseline. |
| S2 | Bi-encoder | Dense retrieval: sentence-transformers/all-MiniLM-L6-v2 (main run); intfloat/e5-small-v2 (fallback). |
| S3 | Cross-encoder | cross-encoder/ms-marco-MiniLM-L-6-v2 reranking the top-100 of S2 (S1 as ablation). |
| S4 | Hybrid (RRF) | Reciprocal Rank Fusion of BM25 and bi-encoder rankings. |

**Rerankers:** applied to the top-50 of each system. (1) **Story cap:** keep one article per story group, detected online in the top-50 by MinHash over title + first ~800 characters. (2) **Outlet cap:** at most 1–2 articles per url_hostname. (3) **Earliness prior:** a separate time-aware variant that boosts the earliest article of each online story group by publish_date (fallback warc_date when implausible), with a weight tuned on the pilot.

**Leakage controls:** retrievers use text only; cluster_id, text_xxhash64 and qrels are never passed to the main systems. The cluster-ID oracle story cap is reported separately as an upper-bound ablation. Rerankers are tuned on the 10 pilot queries only.

## 4. Evaluation Procedure

| Metric | What it captures |
|---|---|
| nDCG@10 | Standard relevance (blind to repetition). |
| α-nDCG@10 (α=0.5) | Coverage of distinct story clusters. |
| ClusterRecall@10 | Share of relevant story clusters shown. |
| Redundancy@10 | Repeat rate in the top-10. |
| Earliest@10 | Whether the first report is surfaced (judged earliest; metrics may use qrels). |
| Latency | Added milliseconds per query for each reranker. |

**Statistics:** BM25 is the primary baseline against S2–S4; each reranker is compared with its un-reranked system. We use paired t-tests on per-query differences [5], because with 50 queries the mean difference is approximately normal whereas Wilcoxon can inflate Type I error, together with bootstrap confidence intervals and Holm correction; key results are confirmed with a paired randomization test. Per-query scatter plots and a split by number of relevant stories (e.g., 6 distinct stories vs. 1 wire story) show when the relevance–coverage trade is free.

**Validation:** Cohen's κ on the 20% double-judged clusters and a manual cluster-precision check on a random sample. Latency is the median of 5 runs on the same machine (CPU and GPU reported separately).

## 5. Timeline

| Weeks | Dates | Deliverable | Owner |
|---|---|---|---|
| 1–2 | 10.11 – 23.11 | Clustering + candidate generation; write 60 queries; pilot judgments; BM25 + dense running. | Uju / Mehdi |
| 3–4 | 24.11 – 07.12 | Full annotation; cross-encoder + hybrid; first results table; validation (κ, cluster precision); scope freeze. | Mehdi / All |
| 5–6 | 08.12 – 21.12 | W5: rerankers + ablation (oracle vs. online story cap). W6: midterm week (15.12), buffer. | Olaniyi |
| 7–8 | 22.12 – 10.01 | Moderator analysis, latency plots, final metrics; README, reproducibility, slides, submission. | Olaniyi / All |

*Risk and fallback:* if annotation runs late, we cut to 40 evaluation queries with 10 candidates each and drop the cross-encoder; judgments stay human.

## 6. Resources

| Resource | Estimate |
|---|---|
| Storage | Download ~700 MB (24 parquet shards); processed corpus ~400 MB; embeddings ~262 MB (fp32). |
| Memory / tokens | ~1.1 TB RAM available; per-month BM25 index < 1 GB. No LLM/API tokens needed. |
| GPU / CPU | 4× L40S on the shared lab server: embeddings in 1–3 min. CPU fallback: embeddings 30–60 min, reranking ~10 min. |
| Person time | ~145 h (3 people × ~6 h/week × 8 weeks): ~15 h annotation (≈5 h each, ~900 decisions) + ~130 h coding/analysis. |

*Fallback:* if inference is too slow on the cluster, we drop the cross-encoder and use BM25 + bi-encoder + rerankers.

## 7. Limitations

One crawl batch per month is not a random sample of that month; publish_date is noisy for ~6% of articles; near-duplicate clustering can miss edited copies (the same_as merge mitigates this); and judgments are pooled, so unjudged clusters are assumed non-relevant. The validation study (κ, cluster precision) quantifies these risks.

## References


[1] Clarke et al., “Novelty and Diversity in Information Retrieval Evaluation”, SIGIR 2008. — *Justifies α-nDCG for penalizing redundant results.*

[2] Lazzaroni, Lasser, Solovev, “Infini-News: Efficiently Queryable Access to 1.3 Billion Processed Common Crawl News Articles”, arXiv:2605.18337, 2026. — *Provided the corpus and the metadata schema used for filtering.*

[3] Xu & Yu, “Demand-Calibrated Facet Diversification with Deployable Soft Quotas for E-Commerce Search”, SIGIR 2026, p. 2129, doi:10.1145/3805712.3809638. — *Informed the story/outlet caps as a deployable diversification mechanism.*

[4] Robertson & Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond”, *Foundations and Trends in Information Retrieval*, 2009. — *The baseline retrieval algorithm.*

[5] Urbano, “Stop Using the Wilcoxon Test: Myth, Misconception and Misuse in IR Research”, SIGIR 2026, p. 2758, doi:10.1145/3805712.3808540 (arXiv:2604.25349). — *Led us to use the paired t-test instead of Wilcoxon.*